

UNIFIED MENTOR

NASSAU CANDY SHIPPING INTELLIGENCE

Factory-to-Customer Shipping Network & Route Analytics

PROJECT RESEARCH REPORT

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INSTITUTION	D. Y. Patil College of Engineering and Technology, Kolhapur
PROJECT	Data Science Internship Project
TECHNOLOGY	Python • Pandas • Plotly • Streamlit

Executive Snapshot

Shipments	Units	Sales	Gross Profit	Margin
10,194	38,654	\$141,783.63	\$93,442.80	65.9%

This project develops an interactive shipping-intelligence solution for Nassau Candy Distributor using 10,194 shipment records and 18 source attributes. The workflow combines data validation, factory mapping, route construction, descriptive statistics, business intelligence and interactive visualization. The resulting Streamlit command center provides a consolidated view of shipment volume, units, sales, gross profit, destination concentration, factory contribution, route activity, shipping-mode mix and product demand. The dataset records 38,654 units, \$141,783.63 in sales and \$93,442.80 in gross profit, with an overall gross-profit margin of 65.9%. During validation, the supplied Order Date and Ship Date fields were found to contain a systematic multi-year offset. Raw date differences are therefore treated as a data-quality diagnostic rather than verified transit-time or delay performance.

Management Takeaways

- California is the largest destination by shipment count: 2,001 shipments (19.63% of all shipments).
- Lot's O' Nuts is the largest factory contributor: 5,692 shipments (55.8%).
- Standard Class is the dominant shipping mode: 6,120 shipments (60.0%).
- Wonka Bar - Milk Chocolate is the highest-volume product: 2,137 shipments and 8,267 units.
- Lot's O' Nuts → California is the largest observed route: 1,125 shipments and \$10,401.00 gross profit.
- Raw shipment-date differences are not used as verified transit-time KPIs because the source dates contain a systematic multi-year offset.

1. Introduction & Business Context

Distribution datasets combine operational and commercial signals: orders, products, factories, customers, geography, shipping modes, revenue, cost and profitability. Business-intelligence methods can transform these records into decision-support views. Supply-chain research identifies data science and big-data analytics as important tools for decision-making across strategic, tactical and operational levels.

This project applies a descriptive analytics and business-intelligence workflow to Nassau Candy Distributor shipment data. The objective is to create a reliable analytical layer that makes network concentration, route activity, product demand and commercial performance easier to inspect.

2. Problem Definition

The raw shipment dataset contains many dimensions but does not provide a single decision-oriented interface. The project addresses two linked problems: converting transactional shipment records into interpretable business intelligence, and preventing invalid operational conclusions when source fields fail basic validation.

3. Objectives & Research Questions

- Measure shipment, unit, sales, cost and gross-profit performance.
- Identify destination and factory concentration.
- Construct factory-to-state routes and compare shipment and commercial contribution.
- Assess shipping-mode and regional shipment mix.
- Identify high-volume products and connect demand with commercial measures.
- Provide interactive filters and shipment-level drill-down.
- Surface data-quality limitations before presenting operational KPIs.

Research questions

- Which destination states contribute the largest share of shipment activity?
- Which factories contribute most strongly to the observed distribution network?

- Which factory-to-state routes have the highest shipment concentration?
- Which products and shipping modes dominate observed activity?
- What data-quality constraints must be resolved before using dates for transit-time or delay analysis?

4. Dataset & Data Dictionary

The supplied analytical dataset contains 10,194 records and 27 prepared columns. The original business attributes include 18 source fields; additional columns were derived for analytical use.

Category	Fields	Purpose
Identity	Row ID, Order ID, Customer ID	Record and customer identifiers
Time	Order Date, Ship Date, Order Month, Ship Month	Temporal fields; date consistency requires validation
Operations	Ship Mode, Factory, Route	Operational network dimensions
Geography	Country/Region, City, State/Province, Postal Code, Division, Region	Destination and regional dimensions
Product	Product ID, Product Name	Product-level analysis
Commercial	Sales, Units, Gross Profit, Cost, Profit Margin %	Financial and volume measures
Derived QA	Shipping Lead Time, Lead Days Mod 365, Adjusted Lead Time, Lead Time Mod 365	Derived diagnostics; not all are validated real-world KPIs

5. Analytical Methodology

1. Data inspection
2. Data-quality validation
3. Factory mapping
4. Feature/route construction
5. Aggregation and KPI calculation
6. Exploratory analysis
7. Interactive dashboard development
8. Interpretation with explicit analytical boundaries

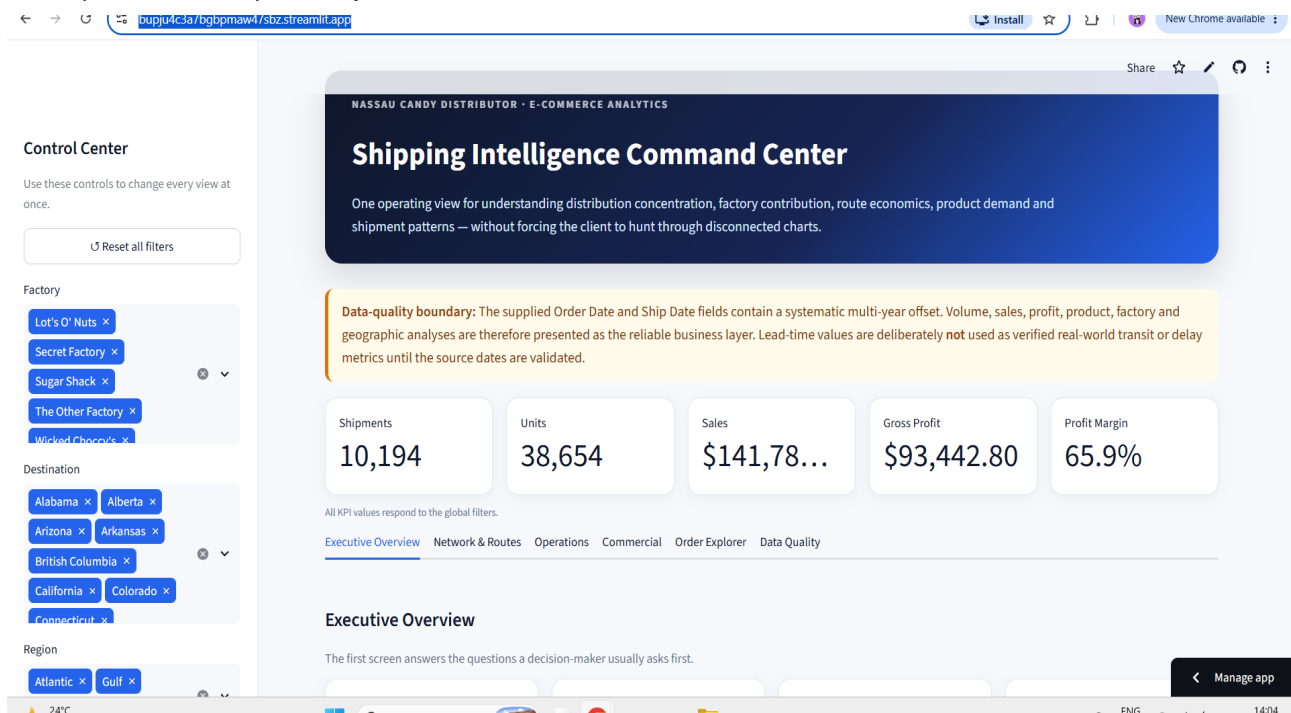


Figure 1. Shipping Intelligence Command Center — executive dashboard view.

6. Data Quality & Validation

A major validation issue was found in the supplied dates. Order Date values range from 02 January 2024 to 31 December 2025, whereas Ship Date values range from 30 June 2026 to 28 June 2030. The raw Shipping Lead Time therefore ranges

from 904 to 1,642 days, with a mean of approximately 1,321 days.

These values are not treated as verified transportation lead times. The dashboard labels the date issue as a data-quality boundary and keeps volume, sales, profit, product, factory and geographic analysis as the primary business layer.

Validation check	Result
Row count	10,194 records
Exact duplicate rows	None identified in the prepared dataset
Missing cells	None identified in the prepared dataset
Order Date range	2024-01-02 to 2025-12-31
Ship Date range	2026-06-30 to 2030-06-28
Raw lead-time range	904–1,642 days
Interpretation boundary	Raw lead time is not a verified transit/delay KPI

7. Exploratory Findings

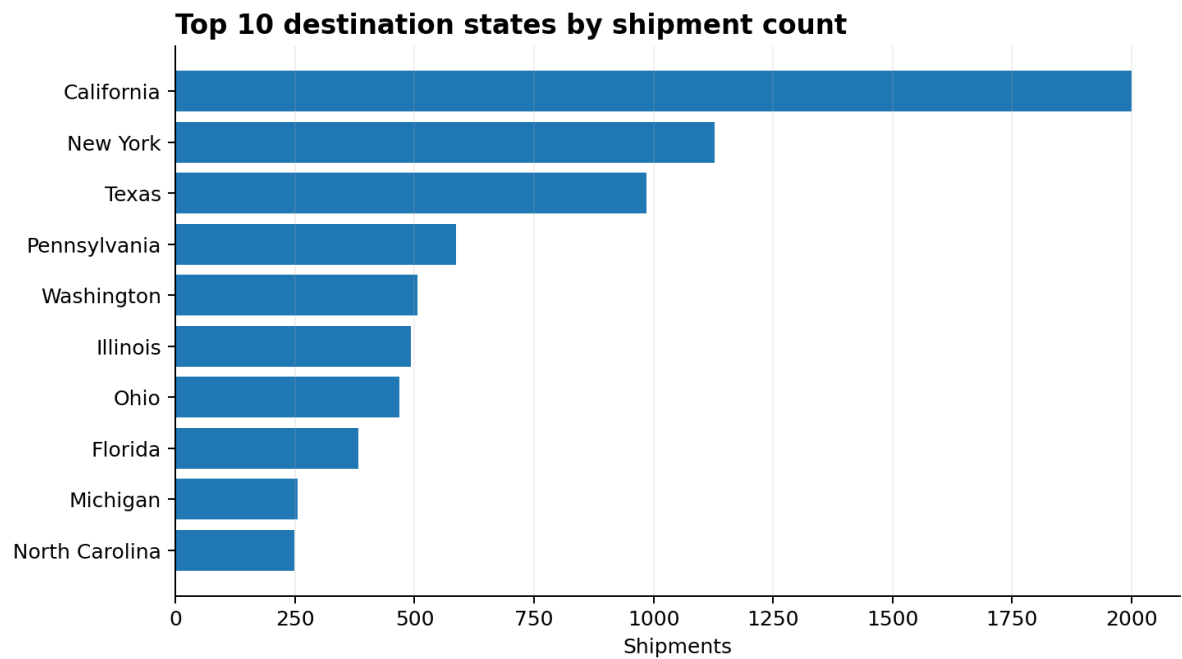


Figure 2. Top destination states by shipment count.

California leads with 2,001 shipments, followed by New York (1,128) and Texas (985).

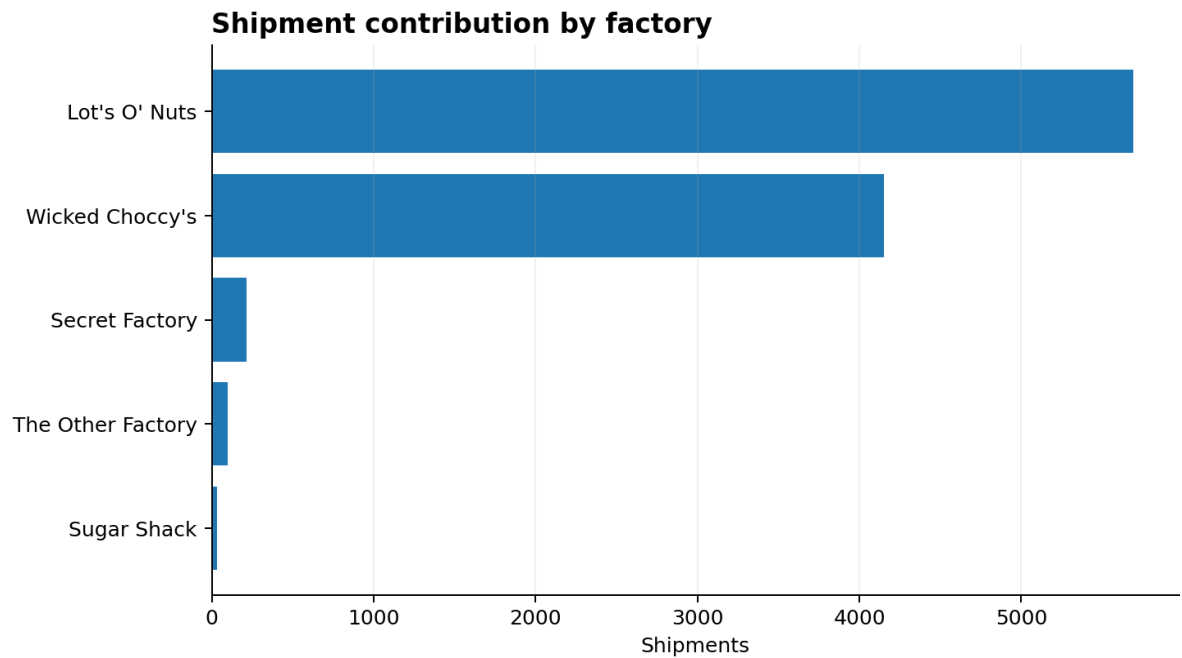


Figure 3. Shipment contribution by factory.

Lot's O' Nuts contributes 5,692 shipments (55.8%), while Wicked Choccy's contributes 4,152 shipments.

8. Route & Network Analysis

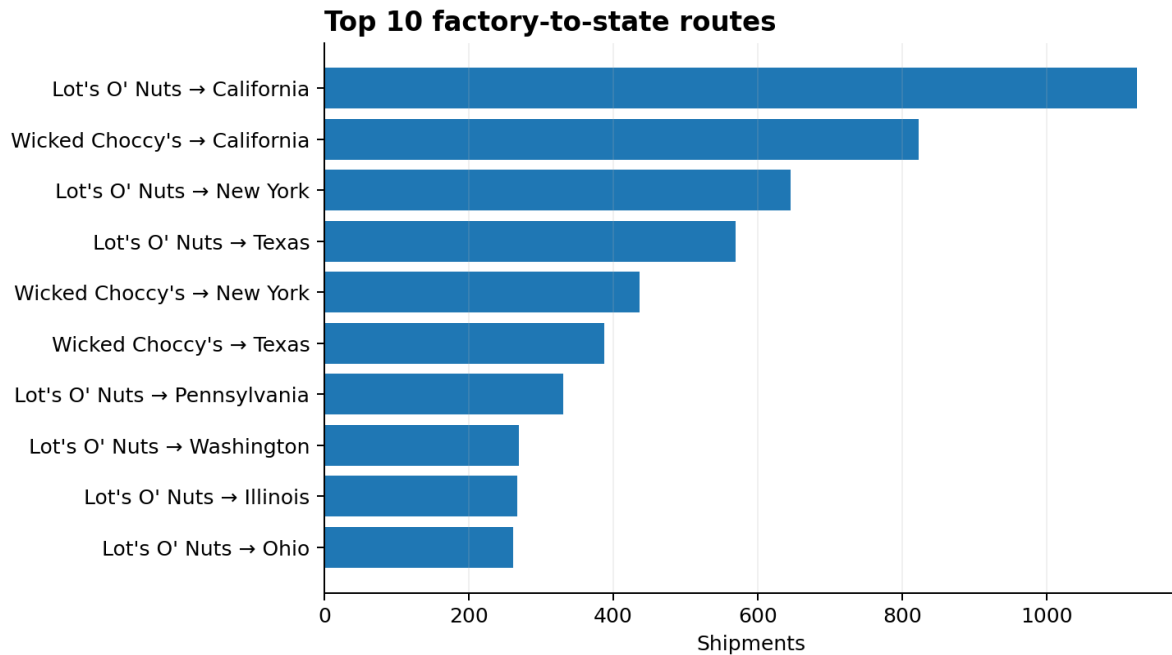


Figure 4. Top factory-to-state routes by shipment count.

The largest route is Lot's O' Nuts → California with 1,125 shipments, 4,221 units, \$15,044.90 in sales and \$10,401.00 in gross profit. Route analysis is based on observed volume and commercial measures rather than unsupported transit-time claims.

9. Commercial & Product Analysis

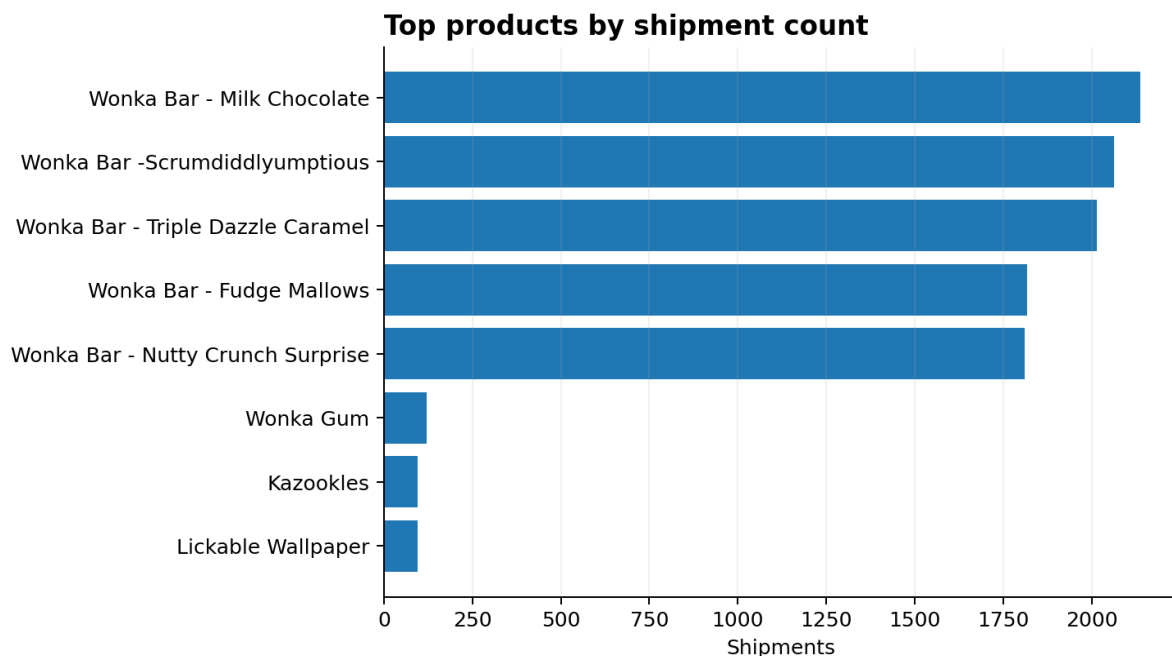


Figure 5. Highest-volume products by shipment count.

Wonka Bar - Milk Chocolate leads with 2,137 shipments and 8,267 units. Wonka Bar - Triple Dazzle Caramel records the highest sales among the top five at \$28,485.00.

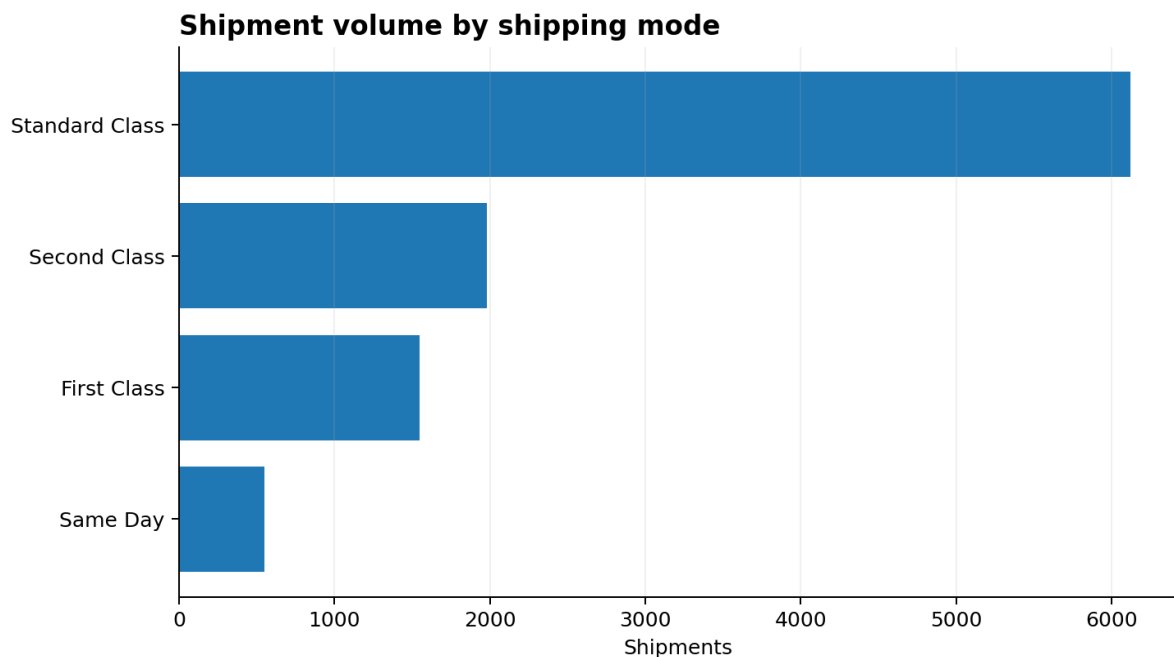


Figure 6. Shipment volume by shipping mode.

Standard Class accounts for 6,120 shipments (60.0%), making it the dominant observed mode.

10. Dashboard Architecture

Module	Purpose
Executive Overview	Global KPIs, concentration indicators and management-level snapshot.
Network & Routes	Destination distribution, route leaderboard, route economics and route drill-down.
Operations	Shipping-mode utilization, regional mix and factory-to-region flow.
Commercial	Product demand, sales concentration and commercial performance.
Order Explorer	Factory/destination filtering and shipment-level records.
Data Quality	Validation findings and explicit boundaries around date-based metrics.

Technology stack: Python, Pandas, Plotly and Streamlit. The dashboard is organized around decision-oriented modules rather than disconnected charts.

11. Decision-Support Recommendations

- Monitor California, New York and Texas as recurring destination concentration indicators.
- Track factory contribution regularly because shipment volume is highly concentrated in the leading factories.
- Use factory-to-state lanes as practical network-monitoring units for volume, units, sales and gross profit.
- Maintain shipping-mode mix as an operational KPI, with Standard Class currently representing the largest observed share.
- Review product concentration together with route concentration to understand where high-volume products create network pressure.
- Do not deploy SLA, delay, transit-time or route-speed rankings until the source date fields are validated.

12. Limitations & Future Scope

The primary limitation is the date inconsistency described above. The project is also descriptive and does not claim a predictive model, causal relationship or optimized factory-assignment solution. Future versions can add verified transit time, delay/service-level metrics, anomaly detection, demand forecasting, route-risk scoring, geospatial distance, transportation-cost estimation and optimization once validated source data is available.

13. Conclusion

The Nassau Candy Shipping Intelligence project transforms 10,194 shipment records into an interactive business-intelligence environment. It provides a consolidated view of 38,654 units, \$141,783.63 in sales and \$93,442.80 in gross profit, while exposing destination, factory, route, product and shipping-mode concentration. Its strongest methodological feature is analytical transparency: validated descriptive findings are separated from date-derived metrics requiring source confirmation.

References

Jahani, H., Jain, R., & Ivanov, D. (2023). Data science and big data analytics: a systematic review of methodologies used in the supply chain and logistics research. *Annals of Operations Research*. DOI: 10.1007/s10479-023-05390-7.

Gonçalves, M., Salgado, C., de Sousa, A., & Teixeira, L. (2025). Data Storytelling and Decision-Making in Seaport Operations: A New Approach Based on Business Intelligence. *Sustainability*, 17(1), 337. DOI: 10.3390/su17010337.

Project Links

Live dashboard: <https://bupju4c3a7bgbpkaw47sbz.streamlit.app/>

GitHub repository: <https://github.com/birajdarsanika717-dyp/Nassau-Candy-Shipping-Intelligence>